Multivariate Data Analysis - MESIO UPC

Correspondence Analysis and Multiple Correspondence Analysis

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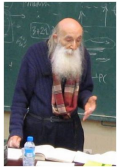
A bit of history



- Benzécri, J.-P. (1973). *L'Analyse des Données*. Volume II. *L'Analyse des Correspondances*.

Goal

Correspondence Analysis focused on obtaining an approximation of a **contingency table** (a two-way frequency table) with the aim of get interpretability and graphical representation. Contingency tables comes from crossing two categorical variables.

Contingency tables are the essential data matrix used for non-parametric test hypothesis such as the χ^2 tests for assessing the independence of two populations. A χ^2 test can reveal that the populations are not independent, then we would want to know how are they associated? This is one of the main goals of CA.

Contingency table (1)

Imagine that our data is composed by two categorical variables A and B with I and J categories respectively, we can represent it the following way if we use an encoding in which the active category is 1 and the rest is 0:

	A_1	A_2	$\dots$	A_I	B_1	B_2	$\dots$	B_J
Individual 1	0	1	$\dots$	0	1	0	$\dots$	0
Individual 2	1	0	$\dots$	0	0	0	$\dots$	1
$\dots$								
Individual n	0	1	$\dots$	0	0	0	$\dots$	1

Contingency table (2)

The next step is to put A as columns and B as rows and each cell will be the number of occurrences in which we have A_i with B_j. This representation is known as the **contingency table**.

	B_1	B_2	$\dots$	B_J
A_1	23	11	$\dots$	10
A_2	3	10	$\dots$	14
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
A_I	55	0	$\dots$	12

Notation

Some notation:

- A: Row variable.
- B: Column variable.
- $\mathbf{N}$: Contingency table.
- r_i: ith row marginal.
- c_j: jth column marginal.
- n: Total count.

Theory

We need the following computations:

- Correspondence matrix:

$$\mathbf{P} = \frac{1}{n}\mathbf{N}$$

- Row vectors:

$$\mathbf{r}_i = \sum_{j=1}^{J} p_{ij} = \sum_{j=1}^{J} \frac{n_{ij}}{n}$$

  for $i = 1, \ldots, I$. Or in matrix form: $\mathbf{r} = \mathbf{P}\mathbf{1}_J$. We build a diagonal matrix with those $\mathbf{D}_r = diag(\mathbf{r})$.

- Column vectors:

$$\mathbf{c}_j = \sum_{i=1}^{I} p_{ij} = \sum_{i=1}^{I} \frac{n_{ij}}{n}$$

  for $j = 1, \ldots, J$. Or in matrix form: $\mathbf{c} = \mathbf{P}^t\mathbf{1}_I$. We build a diagonal matrix with those $\mathbf{D}_c = diag(\mathbf{c})$.

Profiles

Now we are in position of computing the **profile vectors**.

- **Profile**: A vector of non-negative elements that sum up to 1.
- Contingency tables can be converted into a matrix of row profiles or column profiles. Just by dividing each row by its row marginal or each column by its column marginal.
- Row profiles: $\mathbf{R} = \mathbf{D}_r^{-1}\mathbf{P}$
- Column profiles: $\mathbf{C} = \mathbf{D}_c^{-1}\mathbf{P}^t$
- Observe that:

$$\mathbf{r}^t\mathbf{D}_r^{-1}\mathbf{P} = \mathbf{1}^t\mathbf{P} = \mathbf{c}^t$$
$$\mathbf{c}^t\mathbf{D}_c^{-1}\mathbf{P}^t = \mathbf{1}^t\mathbf{P}^t = \mathbf{r}^t$$

Dimensionality

Caused by the nature of the data used in this analysis, we are performing a low rank approximation that will have at most rank $min(I-1, J-1)$.

So it is interesting to obtain both matrices: row profiles and column profiles.

Interpretations

For $\mathbf{R}$, we have that:

- $\mathbf{r}$ is the row mass and $\mathbf{c}$ is the average row profile.
- We are expressing the rows relative to the total number of elements for that category across all the categories of the categorical variable in the columns.

For $\mathbf{C}$, we have that:

- $\mathbf{c}$ is the column mass and $\mathbf{r}$ is the average column profile.
- We are expressing the columns relative to the total number of elements for that category across all the categories of the categorical variable in the rows.

Possibilities

With the contingency table, the profile matrix and the row and column profile matrices, we can:

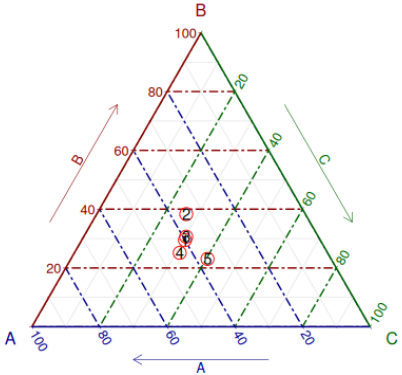
- χ^2 test: Check if the categorical variables are independent or not.
- Compute deviation from each cell in the table.
- If number of dimensions is 3, we can do ternary plots.
- We can do a correspondence analysis.

The ternary plot

We can interpret profiles as compositions (a division of the unity in different parts).

Turns out that there is a nice interpretation for compositions: **The ternary plot**.

Columns profile



The χ^2 distance (1)

As a reminder about the χ^2 statistic formula:

$$\chi^2 = \sum_{i=1}^{I} \sum_{j=1}^{J} \frac{(n_{ij} - e_{ij})^2}{e_{ij}} \tag{1}$$

where $e_{ij} = \frac{n_{i\cdot} n_{\cdot j}}{n} = c_{\cdot j} \frac{r_{i\cdot}}{n} = r_{i\cdot} \frac{c_{\cdot j}}{n}$.

The χ^2 distance (2)

We get the following derivation when considering profiles:

$$\chi^2 = \sum_{i=1}^{I}\sum_{j=1}^{J}\frac{(n_{ij}-e_{ij})^2}{e_{ij}} = \sum_{i=1}^{I}\sum_{j=1}^{J}\frac{(np_{ij}-nr_ic_j)^2}{nr_ic_j} = n\sum_{i=1}^{I}\sum_{j=1}^{J}\frac{(p_{ij}-r_ic_j)^2}{r_ic_j}$$

$$\frac{\chi^2}{n} = \sum_{i=1}^{I}\sum_{j=1}^{J}\frac{(p_{ij}-r_ic_j)^2}{r_ic_j} = \sum_{i=1}^{I}\sum_{j=1}^{J}r_i^2\frac{(\frac{p_{ij}}{r_i}-c_j)^2}{r_ic_j} = \sum_{i=1}^{I}\sum_{j=1}^{J}r_i\frac{(\frac{p_{ij}}{r_i}-c_j)^2}{c_j} = \sum_{i=1}^{I}r_i\sum_{j=1}^{J}\frac{(\frac{p_{ij}}{r_i}-c_j)^2}{c_j}$$

Likewise, for column profiles:

$$\frac{\chi^2}{n} = \sum_{j=1}^{J}c_j\sum_{i=1}^{I}\frac{(\frac{p_{ij}}{c_j}-r_i)^2}{r_i}$$

The χ^2 distance (3)

We obtained the following:

- $\frac{\chi^2}{n}$ is the **total inertia** of the contingency table.
- $\sum_{j=1}^{J} (\frac{p_{ij}}{r_i} - c_j)^2$ is a squared Euclidean distance between profile i and average row profile.
- $\sum_{j=1}^{J} \frac{1}{c_j} (\frac{p_{ij}}{r_i} - c_j)^2$ is a weighted squared Euclidean distance between profile i and average row profile (the χ^2 distance).

Conclusion: Inertia is a weighted average of weighted squared Euclidean distances and a measure of the spread of the profiles *wrt* their average.

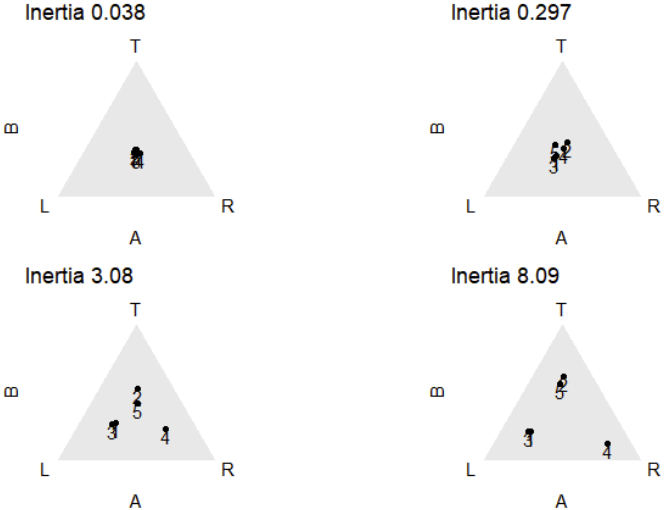
The χ^2 distance (4)

Interpreting the χ^2 distance as a weighted euclidean distance:

- $\sqrt{\sum_j^J \frac{(r_{ij}-r_{i'j})^2}{c_j}} = \|r_i - r_{i'}\|_c$
- $\sqrt{\sum_i^I \frac{(c_{ij}-c_{ij'})^2}{r_i}} = \|c_j - c_{j'}\|_r$
- $\frac{\chi^2}{n} = \sum_i^I r_i \|r_i - c\|_c^2$
- $\frac{\chi^2}{n} = \sum_j^J c_j \|c_j - r\|_r^2$

Inertia

We can visualize the meaning of the inertia using ternary plots:



Limiting situations (1)

- Perfect **independence** between the categorical variables, i.e.: minimal inertia $= 0$ then $\chi^2 = 0$
- Perfect **association** between the categorical variables (are they the same?), i.e.: maximal inertia $= min(I-1, J-1)$

Correspondence Analysis (1)

We are now in position to introduce the **Correspondence Analysis**:

- We want to approximate the profiles in one, two or at most three dimensions.
- The approximation is obtained by a weighted Singular Value Decomposition.
- In this setting, minimization of the errors in the approximation of the profiles is equivalent to maximizing the inertia of the profiles in a k dimensional space (lower or equal to $min(I-1, J-1)$)
- There are different solutions, we will proceed with the use of SVD of the standardized residuals of the contingency table.

Correspondence Analysis (2)

Derivation of the CA solution:

- $\mathbf{E} = \mathbf{r}\mathbf{c}^t$: Expected matrix under the independence hypothesis.
- $\mathbf{R} = \mathbf{P} - \mathbf{E}$: Deviations from independence, those are the **r**esiduals.
- $\mathbf{S} = \mathbf{D}_r^{-\frac{1}{2}}(\mathbf{P} - \mathbf{r}\mathbf{c}^t)\mathbf{D}_c^{-\frac{1}{2}}$: The **s**tandardized residuals.

We will perform SVD on the **S** matrix:

$$\mathbf{D}_r^{-\frac{1}{2}}(\mathbf{P} - \mathbf{r}\mathbf{c}^t)\mathbf{D}_c^{-\frac{1}{2}} = \mathbf{U}\mathbf{D}_\sigma\mathbf{V}^t$$

Standard coordinates:

$$\mathbf{F}_s = \mathbf{D}_r^{-\frac{1}{2}}\mathbf{U} \qquad \mathbf{G}_s = \mathbf{D}_c^{-\frac{1}{2}}\mathbf{V}$$

Principal coordinates:

$$\mathbf{F}_p = \mathbf{D}_r^{-\frac{1}{2}}\mathbf{U}\mathbf{D}_\sigma \qquad \mathbf{G}_s = \mathbf{D}_c^{-\frac{1}{2}}\mathbf{V}\mathbf{D}_\sigma$$

Correspondence Analysis representation (1)

Different ways of representing the solution are possible:

- $\mathbf{F}_s$ with $\mathbf{G}_p$: Biplot of scaled row profiles.
- $\mathbf{F}_p$ with $\mathbf{G}_s$: Biplot of scaled column profiles.
- $\mathbf{D}_r^{-\frac{1}{2}}\mathbf{U}\mathbf{D}^{\frac{1}{2}}$ with $\mathbf{D}_c^{-\frac{1}{2}}\mathbf{V}\mathbf{D}^{\frac{1}{2}}$: Symmetric biplot.
- $\mathbf{F}_p$ with $\mathbf{G}_p$ (Not a biplot).

Correspondence Analysis representation (2)

Observe that $\mathbf{F}_s\mathbf{G}_p^t = \mathbf{D}_r^{-\frac{1}{2}}\mathbf{U}\mathbf{D}_\sigma\mathbf{V}^t\mathbf{D}_c^{-\frac{1}{2}} = (\mathbf{D}_r^{-1}\mathbf{p} - \mathbf{1}\mathbf{c}^t)\mathbf{D}_c^{-1}$ and that $\mathbf{G}_s\mathbf{F}_p^t = \mathbf{D}_c^{-\frac{1}{2}}\mathbf{V}\mathbf{D}_\sigma\mathbf{U}^t\mathbf{D}_r^{-\frac{1}{2}} = (\mathbf{D}_c^{-1}\mathbf{p} - \mathbf{1}\mathbf{r}^t)\mathbf{D}_r^{-1}$, then, for a better representation, the biplot column vectors can be scaled by the pre-multiplication of $\mathbf{G}_p$ by $\mathbf{D}_c^{-\frac{1}{2}}$ giving

$$\mathbf{F}_s\mathbf{D}_c^{\frac{1}{2}}\mathbf{G}_p^t = (\mathbf{D}_r^{-1}\mathbf{p} - \mathbf{1}\mathbf{c}^t)\mathbf{D}_c^{-1}\mathbf{D}_c^{\frac{1}{2}} = (\mathbf{D}_r^{-1}\mathbf{p} - \mathbf{1}\mathbf{c}^t)\mathbf{D}_c^{-\frac{1}{2}}$$

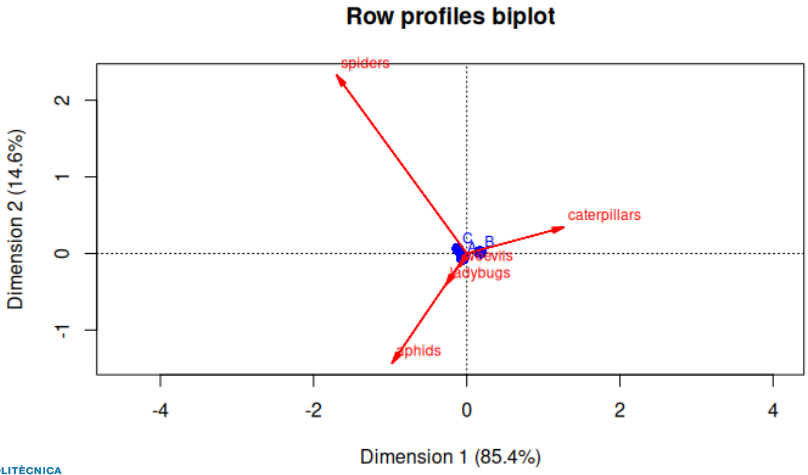
The last equation is known as the "**contribution**" biplot.

Row profiles biplot (1)

Categories in rows as points and the columns as arrows (vectors).

- Coordinates:
  - Rows: Principal coordinates (preserving the chi-square distance between rows).
  - Columns: Standard coordinates (unscaled). Represents directions.
- Interpretations:
  - Row proximity: Closer poitns have similar row profiles.
  - Column vectors: Projecting a row point onto a column vector approximates the value in the row profile. Association of columns in similar direction: similar row patterns.
  - Origin: Is the average profile. Deviations indicate uniqueness.

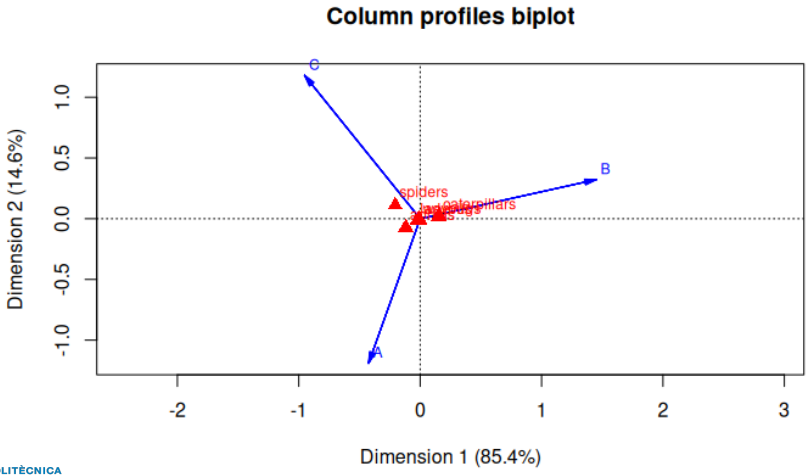
Row profiles biplot (2)



Column profiles biplot (1)

Categories in columns as points and the rows as arrows (vectors). Coordinates and interpretations are analogous to those of the Row profiles biplot but interchanging rows by columns.

Column profiles biplot (2)

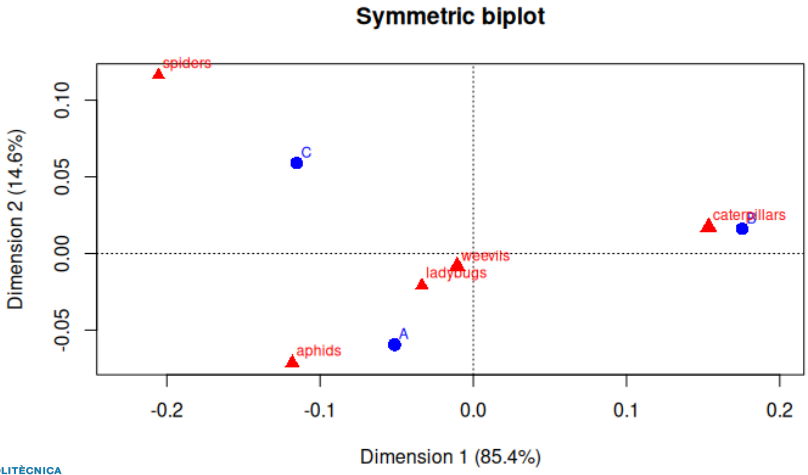


Symmetric biplot (1)

All variables are represented as points. The representation between rows and columns is balanced.

- **Coordinates**: Both rows and columns are scaled by square roots of singular values.
- **Interpretation**: No pure chi-squared distances. We can get association of rows and columns through geometry using angles. Gives an holistic interpretation of the relationships between rows and columns.

Symmetric biplot (2)



Contribution biplot

Emphasis on showing the **contributions** of rows/columns to the inertia of axes.

- Coordinates: Points scaled by contributions to each axis. High contribution points are stretched along axes they influence.
- Interpretation: Points far apart from the origin are major contributions to that dimension's inertia. Helps to identify structure of the data (outliers, influential categories, etc.).

Transitions

As there are two profiles, we can obtain one of the solutions from the other through the transition relationships:

- $\mathbf{F}_p = \mathbf{D}_r^{-1}\mathbf{P}\mathbf{G}_s$
- $\mathbf{G}_p = \mathbf{D}_c^{-1}\mathbf{P}^t\mathbf{F}_s$
- Principal coordinates of the rows are weighted averages of standard coordinates of the columns.
- These can be used for calculate the coordinates of supplementary points.

Supplementary data

- Supplementary points (added at posteriori) or inactive points during the solution's construction (rows or columns) are usually **data collected under different conditions**.
- Their representation on the biplot in a **posterior analysis** can be helpful for **interpretation**.
- Supplementary points can be situated in CA biplots by expressing them as profiles and using the transition relationships.

Decomposition of the inertia

As happened with other analysis, the total inertia is decomposed through the value of the obtained eigenvalues. The relation between the total sum of the eigenvalues and the total inertia is given by the following formula:

$$\frac{\chi^2}{n} = \sum_i \lambda_i$$

Contributions to the inertia (1)

Similar to the PCA case, inertia can come from different sources: the contribution made by the principal components, the contribution from the variables and the contribution from the individuals. In CA a similar decomposition is possible by different sources:

- Contribution made by dimensions (principal axis).
- Contribution made by the row variable.
- Contribution made by the column variable.
- Contribution made by the individual cells of the table.

Each source of inertia provides a different or a more detailed interpretation of the data.

Contributions to the inertia (2)

- **Row inertias**: $\frac{\chi^2}{n} = \sum_{i=1}^{I} r_i \sum_{j=1}^{J} \frac{(\frac{p_{ij}}{r_i} - c_j)^2}{c_j}$
- **Column inertias**: $\frac{\chi^2}{n} = \sum_{j=1}^{J} c_j \sum_{i=1}^{I} \frac{(\frac{p_{ij}}{c_j} - r_i)^2}{r_i}$
- **Decomposition**:

$$\frac{\chi^2}{n} = \sum_{j=1}^{J} \sum_{i=1}^{I} \frac{(p_{ij} - r_i c_j)^2}{r_i c_j} = tr(\mathbf{D}_r^{-1}(\mathbf{P} - \mathbf{r}\mathbf{c}^t)\mathbf{D}_c^{-1}(\mathbf{P} - \mathbf{r}\mathbf{c}^t)^t) = tr(\mathbf{D}^2)$$

Contributions to the inertia (3)

- Squared singular values (eigenvalues) are called **principal inertias** and constitute the contribution of each dimension to the total inertia.
- The inertias of each row/column can be decomposed into contributions made by the principal axis. This allows one to judge how much of the inertia of each row/column is accounted for by each axis, and to compute goodness-of-fit statistics for each point.
- Points contributions can be interpreted as:
  - **Square cosines**: Point representation.
  - **Contributions**: Contribution to the inertia captured by the principal axis.

  Note: The concepts of square cosines and contributions are analogous to those in PCA.

Multiple Correspondence Analysis

A bit of history

- Ludovic Lebart, main promoter of the Multiple Correspondence Analysis.

Context

Multiple correspondence analysis (MCA) is a generalization of CA. It performs a joint study of multiple categorical random variables.

This kind of analysis is very useful for:

- Survey analysis.
- Species distribution and habitat characteristics.
- Genetic variation and gene expression.

Example data (1)

During this presentation we will use a simulated example survey with 4 questions made to 150 respondents:

- **Age**: Age group with values: "18-25", "26-40", "41-60", "60+"
- **Transportation mode**: Values "Transit", "Bike", "Car"
- **Dining Out Frequency**: Values "Rarely", "Weekly", "Daily"
- **Coffee Preference**: Values "Black", "Milk", "Tea"

Example data (2)

A look at the data:

n	Age	Transport	Dining	Coffee
1	41-60	Car	Daily	Milk
2	18-25	Transit	Rarely	Black
3	26-40	Bike	Weekly	Black
4	26-40	Bike	Daily	Black
5	26-40	Bike	Rarely	Black
6	41-60	Car	Weekly	Tea
⋮	⋮	⋮	⋮	⋮

MCA

MCA consists in applying CA to a two specific types of matrices:

- The **indicator** matrix.
- The **Burt** matrix.

Indicator matrix (1)

An indicator matrix $\mathbf{Z}$ arise for having all categorical variables of a dataset in a one-hot encoding format, hence the name.

	Original data								$\mathbf{Z}_1$					$\mathbf{Z}_2$						
	A	B	C	D	E	F	G	H	1	2	3	4	5	1	2	3	4	5	1	2
1	2	4	3	3	4	1	4	1	0	1	0	0	0	0	0	0	1	0	0	0
2	3	2	2	3	4	1	3	2	0	0	1	0	0	0	1	0	0	0	0	1
3	2	4	4	4	2	2	5	1	0	1	0	0	0	0	0	0	1	0	0	0
4	1	3	2	3	2	4	4	2	1	0	0	0	0	0	0	1	0	0	0	1
5	2	4	2	3	4	4	5	1	0	1	0	0	0	0	0	0	1	0	0	1
6	2	4	2	2	4	2	4	1	0	1	0	0	0	0	0	0	1	0	0	1
7	2	2	2	4	4	2	4	1	0	1	0	0	0	0	1	0	0	0	0	1
8	4	2	2	2	4	1	5	1	0	0	0	1	0	0	1	0	0	0	0	1
9	4	2	2	4	4	2	4	1	0	0	0	1	0	0	1	0	0	0	0	1
10	3	3	3	3	2	2	4	2	0	0	1	0	0	0	0	1	0	0	0	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Indicator matrix (2)

Our data now has the following form:

$$\mathbf{Z} = [\mathbf{N}_1, \mathbf{N}_2, \ldots, \mathbf{N}_q]$$

$\mathbf{Z}$ is of order $(n \times J)$

Remember that in CA we had $\mathbf{N} = [\mathbf{A}, \mathbf{B}]$

Indicator matrix (3)

- Q: number of categorical variables.
- J_q: number of categories for variable q.
- J: Total number of categories.
- $\mathcal{I}(\cdot)$: Inertia.

$$J = \sum_{q=1}^{Q} J_q$$

$$\mathcal{I}(\mathbf{N}_q) = J_q - 1$$

$$\mathcal{I}(\mathbf{Z}) = \frac{\sum_q \mathcal{I}(\mathbf{N}_q)}{Q} = \frac{J - Q}{Q}$$

The last formula gives us that the total inertia on the Indicator matrix is the mean of the inertias in all subtables. Consequently, the **Inertia per dimension** is $\frac{1}{Q}$.

Indicator matrix (4)

A consequence of the previous formula is that the **inertia** in the **Indicator matrix** does not depend on the data, it **depends on** its **structure** (number of categorical variables and total number of categories).

Indicator matrix (5)

Indicator table of the example data:

	Age Group				Transportation			Dining			
	18-25	26-40	41-60	60+	Bike	Car	Transit	Daily	Rarely	Weekly	...
1	0	0	1	0	0	1	0	1	0	0	...
2	1	0	0	0	0	0	1	0	1	0	...
3	0	1	0	0	1	0	0	0	0	1	...
4	0	1	0	0	1	0	0	1	0	0	...
5	0	1	0	0	1	0	0	0	1	0	...
6	0	0	1	0	0	1	0	0	0	1	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	...

Indicator matrix (6)

Total inerta computation:

- **Total number of questions** (Q): 4
- **Total number of categories** (J):
  - **Age**: 4 categories.
  - **Transport**: 3 categories.
  - **Dinning**: 3 categories.
  - **Coffee**: 3 categories

  The total is 13.

The total inertia is then:

$$\frac{J-Q}{Q} = \frac{13-4}{4} = 2.25$$

Burt matrix

A Burt matrix $\mathbf{B}$ is a symmetric $J \times J$ matrix containing all possible two-way combinations of tables between Q categorical variables.

$$\mathbf{B} = \mathbf{Z}^t\mathbf{Z} = \begin{bmatrix} \mathbf{N}_1^t\mathbf{N}_1 & \mathbf{N}_1^t\mathbf{N}_2 & \dots & \mathbf{N}_1^t\mathbf{N}_q \\ \mathbf{N}_2^t\mathbf{N}_1 & \mathbf{N}_2^t\mathbf{N}_2 & \dots & \mathbf{N}_2^t\mathbf{N}_q \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{N}_q^t\mathbf{N}_1 & \mathbf{N}_q^t\mathbf{N}_2 & \dots & \mathbf{N}_q^t\mathbf{N}_q \end{bmatrix}$$

Burt matrix (1)

Burt matrix of our example data:

	Age Group				Transportation			Dining			
	18-25	26-40	41-60	60+	Bike	Car	Transit	Daily	Rarely	Weekly	Black
18-25	34.00	0.00	0.00	0.00	15.00	2.00	17.00	18.00	10.00	6.00	7.00
26-40	0.00	49.00	0.00	0.00	18.00	7.00	24.00	14.00	17.00	18.00	20.00
41-60	0.00	0.00	45.00	0.00	4.00	36.00	5.00	17.00	13.00	15.00	15.00
60+	0.00	0.00	0.00	22.00	5.00	16.00	1.00	9.00	7.00	6.00	7.00
Bike	15.00	18.00	4.00	5.00	42.00	0.00	0.00	19.00	11.00	12.00	17.00
Car	2.00	7.00	36.00	16.00	0.00	61.00	0.00	22.00	20.00	19.00	21.00
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Burt matrix (2)

Considerations about the inertia in a Burt matrix:

- If all two-way tables have the same margins (no missing data on any of the variables) then the inertia of the Burt matrix is the average of the inertias of all two-way tables.
- Otherwise (if tables have missing data) this will be approximately true.

- The matrices on the diagonal of the Burt matrix have maximal inertia, $J_q - 1$ each. This inertia should not contribute to the analysis, then:
  - Total inertia of all subtables in the Burt matrix: $Q^2\mathcal{I}(\mathbf{B})$
  - Total inertia of all subtables in the diagonal: $\sum_{q=1}^{Q}(J_q - 1) = J - Q$
  - Total inertia of all subtables in the off-diagonal: $Q^2\mathcal{I}(\mathbf{B}) - (J - Q)$
  - Then the (average) inertia in the off-diagonal part of the Burt matrix is:

$$\mathcal{I}^{adj}(\mathbf{B}) = \frac{1}{Q^2 - Q}\left(Q^2\mathcal{I}(\mathbf{B}) - (J - Q)\right)$$

Burt matrix (3)

The **unadjusted inertia of the Burt matrix** can be calculated if one observes that MCA on a Burt matrix is equivalent to perform CA on sub-matrices that are contingency tables, so the inertia is the sum of the individual inertias. Takin into account that the diagonal elements are the rows and columns marginals of each contingency sub-table (they are equal because the matrix is symmetric). Then we can compute the unadjusted inertia as:

$$\mathcal{I}(\mathbf{B}) = \frac{1}{Q^2} \sum_{i=1}^{I} \sum_{j=1}^{J} \frac{b_{ij}^2}{r_i c_j} - 1$$

where Q is the total number of questions (variables), b_{ij} is the entry ij of the Burt table and r_i and c_j are the row and column marginal for categorical variable i and j (e.g.: number of people who chose category i and category j).

Burt matrix (4)

The total inertia computed from the Burt matrix in our example data is:

$$\mathcal{I}^{adj}(\mathbf{B}) = \frac{1}{Q^2 - Q}\left(Q^2\mathcal{I}(\mathbf{B}) - (J - Q)\right) = \frac{1}{16 - 4}\left(16 * 0.6367 - (13 - 4)\right) = 0.099$$

Inertia discrepancy

The total inertia differs between the Indicator matrix and the Burt matrix. **Why?**

The computed **Indicator matrix inertia** is artificially inflated because it is primarily a function of the matrix structure (the number of categorical variables and their levels) rather than the associations within the data. In contrast, the **Burt matrix adjusted inertia** filters out this structural redundancy, providing a more accurate measure of the true variance and relationships present in the dataset.

MCA

We can obtain the MCA solution by different means:

- SVD on indicator standardized matrix.
- Eigendecomposition on Burt standardized matrix.

Steps in both approaches are analogue to those in CA.

MCA: Indicator matrix (1)

The steps to compute the MCA solution from an **Indicator** matrix are:

1. Compute the **Probability** matrix:

$$P = \frac{1}{NQ} Z$$

   where N is the number of rows and Q is the number of columns.

2. Compute the **Standardized residuals** matrix:

$$S = Dr^{-\frac{1}{2}}(P - rc^t)D_c^{-\frac{1}{2}}$$

   where r are the row sums of P, c are the column sums of P, $D_r = diag(r)$ and $D_c = diag(c)$.

3. Perform **SVD** on S:

$$S = UD_\sigma V^t$$

MCA: Indicator matrix (2)

SVD on S gives us:

- **Square root of the Inertias**: As the singular values, denoted σ_{ii}. Inertias are then computed as $\lambda_i = \sigma_{ii}^2$
- **Principal coordinates for categories**: The column space (categories), computed as:

$$F = D_r^{-\frac{1}{2}} U D_\sigma$$

- **Principal coordinates for individuals**: The eigenvectors in U, describing the row space (individuals).

$$G = D_c^{-\frac{1}{2}} V D_\sigma$$

MCA: Burt matrix (2)

The steps to compute the MCA solution from a **Burt** matrix ($B = Z^t Z$) are:

1. Compute **Probability** matrix:

$$P_B = \frac{1}{NQ^2} B$$

   where N is the number of individuals and Q is the number of columns. Observe that NQ^2 is the sum of all entries in B.

2. Compute the **Standardized residuals** matrix:

$$S_B = D_r^{-\frac{1}{2}} (P_B - rc^t) D_c^{-\frac{1}{2}}$$

   where r are the row sums of P, c are the column sums of P, $D_r = diag(r)$ and $D_c = diag(c)$. Since B is symmetric, then $r = c$

3. **Eigendecomposition** on S_B:

$$S_B = V D_\lambda V^t$$

MCA: Burt matrix (1)

Eigendecomposition on S gives us:

- **Inertias**: As the eigenvalues, denoted λ_i. Remember that $\lambda_i = \sigma_{ii}^2$.
- **Principal coordinates for categories**: The eigenvectors in V, describing the column space (categories).

$$G = D_c^{-\frac{1}{2}} V D_\lambda^{\frac{1}{2}}$$

  Note that $D_\sigma = D_\lambda^{\frac{1}{2}}$
- **Principal coordinates for individuals**: These are not given, but can be computed using the **transition formula**:

$$F = \frac{1}{Q} Z G D_\lambda^{-\frac{1}{4}}$$

MCA: Z or B? (1)

Differences between performing MCA with $\mathbf{Z}$ or with $\mathbf{B}$:

- Standard coordinates of MCA are the same.
- Eigenvalues of $\mathbf{B}$ are the squares of the eigenvalues obtained in the analysis of $\mathbf{Z}$.
- Percentages of explained inertia are therefore higher when using $\mathbf{B}$.
- Principal coordinates of MCA with $\mathbf{B}$ are shrunk *wrt* MCA with $\mathbf{Z}$.
- As the Burt matrix is an square matrix, an spectral decomposition can be directly used and the obtained eigenvalues give a better approximation of the inertia explained by the factors when compared to those obtained from $\mathbf{Z}$.

MCA: Categories vs Individuals

MCA gives us two interpretable spaces:

- **The categories space**, G: It will show the structure of our data and provide interpretations about possible questions in a survey. This one is the most used case.
- **The individuals space**, F: Use it in case of looking to segment the respondents, identify outliers, color-code groups in the biplots, etc.

MCA: Inertia Problems (1)

Either if MCA is computed from Indicator or from Burt, the total inertia (sum of eigenvalues) will be higher than the total inertia from the Indicator or Burt matrix. This is caused by:

- **MCA Indicator**: The structure of the Indicator matrix (mutually exclusivity of the answers in a question) introduces structure that causes an **implicit** inflation of the λ_i.
- **MCA Burt**: The diagonal elements of the Burt matrix cause an explicit inflation of the λ_i

So whatever the method, one must always perform an adjustment, **the Benzecri correction**.

MCA: Inertia Problems (2)

Since the Burt Matrix inflates the inertias, one should always take into account the following relationship of scales for a given singular value σ_{ii} its inertia is:

- **Indicator matrix**: $\lambda_i = \sigma_{ii}^2$
- **Burt matrix**: $\lambda_i = \sigma_{ii}^4$

This implies that the **Benzecri correction** depends on the matrix used for the analysis:

- **Indicator matrix**:

$$\lambda_k^{adj} = \left(\frac{Q}{Q-1}\right)^2 \left(\lambda_k - \frac{1}{Q}\right)^2 \quad \text{for } \lambda_k > \frac{1}{Q}$$

- **Burt matrix**:

$$\lambda_k^{adj} = \left(\frac{Q}{Q-1}\right)^2 \left(\sqrt{\lambda_k} - \frac{1}{Q}\right)^2 \quad \text{for } \sqrt{\lambda_k} > \frac{1}{Q}$$

MCA: Inertia Problems (3)

In our survey data example, the inertia from MCA using the **Indicator matrix** is:

	dim 1	dim 2	dim 3	dim 4	dim 5	dim 6	dim 7	dim 8	dim 9
λ_i	0.42	0.32	0.30	0.27	0.25	0.23	0.21	0.18	0.08
%	18.81	14.11	13.21	12.07	10.95	10.37	9.16	7.83	3.50
cumul.	18.81	32.92	46.13	58.20	69.15	79.52	88.67	96.50	100.00

While the **Indicator matrix adjusted inertia** is:

	dim 1	dim 2	dim 3	dim 4
λ_i	0.0533	0.0081	0.0040	0.0008
%	80.5145	12.2382	5.9856	1.2617
cumul.	0.0533	0.0614	0.0653	0.0662

MCA: Inertia Problems (4)

In our survey data example, the inertia from MCA using the **Burt matrix** is:

	dim 1	dim 2	dim 3	dim 4	dim 5	dim 6	dim 7	dim 8	dim 9
λ_i	0.18	0.10	0.09	0.07	0.06	0.05	0.04	0.03	0.01
%	28.12	15.83	13.87	11.59	9.53	8.55	6.66	4.87	0.97
cumul.	28.12	43.95	57.82	69.41	78.94	87.49	94.15	99.03	100.00

While the **Burt matrix adjusted inertia** is:

	dim 1	dim 2	dim 3	dim 4
λ_i	0.0533	0.0081	0.0040	0.0008
%	80.5145	12.2382	5.9856	1.2617
cumul.	0.0533	0.0614	0.0653	0.0662

MCA: Inertia Problems (5)

Then the total inertia (accumulated sum of eigenvalues) is:

- **Unadjusted inertia**:
  - **Indicator matrix**: 2.25
  - **Burt matrix**: 0.6367
- **Adjusted inertia**:
  - **Indicator matrix**: 0.0662
  - **Burt matrix**: 0.0662

Observe that the **Burt Adjusted Inertia** was 0.099. The discrepancy is due to the filtering of eigenvalues greater than $\frac{1}{Q}$

MCA final solution

- The MCA solution from **Indicator** matrix provides better plots (using the σ_{ii} but inertias need to be obtained from σ_{ii}^2 to obtain good interpretations of the principal axes.
- The MCA solution from **Burt** matrix provides squishy plots but the inertias (adjusted) can provide good interpretations of the principal axes.

The solution will be a combination of both:

1. SVD on Indicator then get G, F and σ_{ii}^2.
2. Adjust inertias (Benzecri correction).
3. Do plots using σ_{ii} and provide interpretations based on the adjusted inertias λ_i^{adj}

Decomposition of the inertia (1)

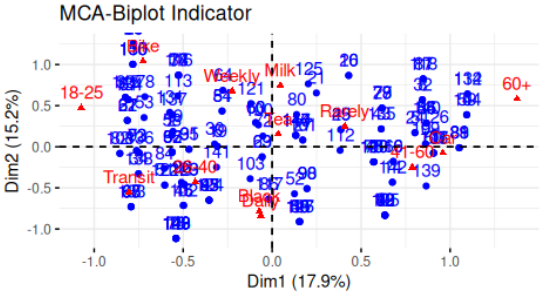
Similarly to PCA and CA, the inertia can be decomposed down to different sources that provide different interpretabilities:

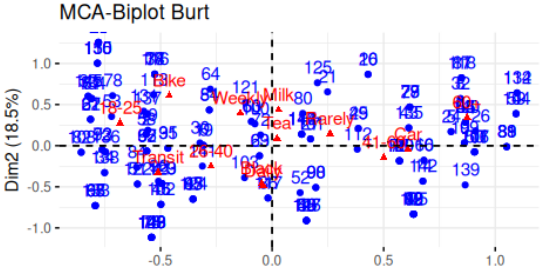
- λ_k inertia of the principal axis k.

Decomposition of the inertia (2)

- **Quality of representation**: Measures how well a specific point is projected onto the axis k (a value near 1 is perfect representation):
  - **For individual** i: $cos^2_{ik} = \frac{f^2_{ik}}{d^2_i}$
  - **For category** j: $cos^2_{jk} = \frac{g^2_{ik}}{d^2_j}$ (d^2 is the point squared geometric distance to the origin)
- **Contributions to the axis formation**: Measures how much a specific point was involved with the creation of the axis k (sum of all contributions on an axis is 100%):
  - **For individual** i: $ctrl_{ik} = \frac{m_i f^2_{ik}}{\lambda_k}$
  - **For category** j: $ctrl_{jl} = \frac{m_j g^2_{jk}}{\lambda_k}$ (m_i and m_j are the respective statistical masses of the individual and the category)

Biplot visual differences





CA and MCA in R

The following libraries allows us to perform correspondence analysis in R:

- **ca**: Package that allows to perform simple and direct CA.
- **FactoMineR**: Automation of CA and MCA with a lot of auxiliary methods for evaluation and plotting. It can be enhanced with **factoextra**.
- **sparseMCA**: MCA for sparse data (data with a lots of zero values), common in indicator matrices.

References

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